

MINI CANDLE LIGHT

Using Old Mobile Phone Battery

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According to the three Rs—reuse, reduce and recycle—we must save our planet from e-waste. Sometimes we have a dead mobile phone whose battery is in good working condition. The circuit given here is used to utilise such batteries for making a mini candle light using bright LEDs.

The circuit is built around MC34063 DC-to-DC converter IC. It is a monolithic controller that consists of internal temperature-compensated

voltage reference, comparator, controlled duty-cycle oscillator with an active current-limit circuit, driver and high-current output switch. This IC was specifically designed to step-down, step-up and for voltage-inverting applications with a minimum number of external components.

Circuit and working

Circuit diagram of the mini candle light using an old mobile phone battery is shown in Fig. 1. It is built around monolithic DC-to-DC converter controller MC34063 (IC1), 100µH inductor (L1), Schottky barrier diode 1N5819 (D1), four 5mm bright white LEDs and a few other components.

Here, IC1 is configured in step-up configuration to drive the bright white LEDs using an old mobile phone battery. The 33-ohm resistor (R4) limits the current flowing through the LEDs. Capacitor C2 determines the operating frequency of IC1, and R3 and R5 determine the output voltage.

Output voltage can be calculated by using the relationship:

$$V_{out} = 1.25 [1 + (R3/R5)]$$

When switch S1 is closed, IC1, inductor L1 and Schottky

PARTS LIST

Semiconductors:	
IC1	- MC34063A DC-to-DC converter
D1	- 1N5819 Schottky diode
LED1-LED4	- 5mm bright white LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 180-ohm
R2	- 0.47-ohm
R3	- 10-kilo-ohm
R4	- 33-ohm
R5	- 1-kilo-ohm
Capacitors:	
C1	- 100µF, 16V electrolytic
C2	- 100pF ceramic disk
C3	- 470µF, 16V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector
S1	- On/off switch
	- 3.7V old mobile phone battery

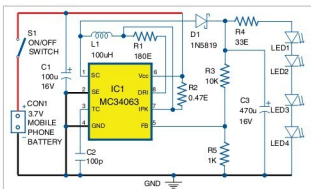


Fig. 1: Circuit diagram of the mini candle light

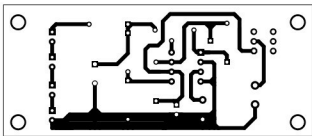


Fig. 2: Actual-size PCB layout of the mini candle light

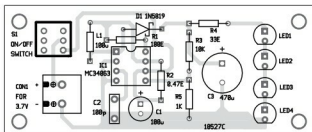


Fig. 3: Component layout of the PCB

barrier diode D1 perform step-up operation and LEDs (LED1 through LED4) glow.

IC 78S40 can be used in place of MC34063. Any Schottky barrier diode works fine in this circuit. By using a low-ESR capacitor, you can increase the efficiency of the circuit.

The circuit can be converted into a small emergency light by adding a charger and a few more components.

Construction and testing

An actual-size, single-side PCB layout for the mini candle light using an old mobile phone battery is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable plastic box.

The circuit works off an old mobile phone's 3.7V battery. It will also, of course, work with any 3.7V DC power supply. **EFY**



A. Samiuddin is an electronics hobbyist. His interests include LED lightings, power electronics, microcontrollers and Arduino programming